

BURAK UZKENT, Ph.D.

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👤 PROFESSIONAL SUMMARY

Principal Member of Technical Staff with **10+ years** of experience developing and deploying large-scale machine learning systems. Specialized in **generative AI**, **computer vision**, **multi-modal and video-language modeling**, and efficient **transformer architectures**. Published **40+ papers** in top-tier venues (CVPR, ICCV, ICLR, NeurIPS, EMNLP, AAAI) across academic research and industrial applications. Proven track record of leading research initiatives at AMD, Amazon, Samsung, and Stanford University.

👜 PROFESSIONAL EXPERIENCE

Principal Member of Technical Staff

AMD

📅 April 2026 – Present

📍 Santa Clara, CA

- ✔ Work on applications of **generative AI** on AMD hardware
- ✔ Develop and evaluate ML systems optimized for AMD accelerators and platforms
- ✔ Hiring for full-time positions at all levels

Machine Learning Scientist

Amazon Prime Video

📅 April 2022 – March 2026

📍 Sunnyvale, CA

- ✔ Led development of **Video LLMs** for advanced video understanding and content moderation in long-form videos
- ✔ Designed and implemented multimodal foundation models for video summarization pipeline
- ✔ Developed transformer-based NLP models for subtitle analysis enabling automated content moderation at scale
- ✔ Published **6 papers** (CVPR, WACV, EMNLP, ECCV Workshop) and filed **2 patents**; 1 additional paper under review

Senior Research Scientist

Samsung Research America

Manager: Hongxia Jin, Ph.D.

📅 November 2020 – April 2022

📍 Mountain View, CA

- ✔ Optimized vision transformer architectures achieving **significant model compression** while maintaining accuracy
- ✔ Developed efficient multi-modal transformers for on-device applications with reduced inference latency
- ✔ Led team of **3 researchers** in developing novel computer vision and multimodal ML models
- ✔ Published **4 papers** in top-tier conferences (CVPR, ICLR, AAAI) and filed **6 patents**

Postdoctoral Fellow

Stanford University, Department of Computer Science
 Advisor: Stefano Ermon, Ph.D.

📅 July 2018 – October 2020
 📍 Stanford, CA

- ✓ Published **15+** papers in top-tier conferences (ICCV, CVPR, ICLR, AAAI, IJCAI, KDD)
- ✓ Developed novel self-supervised and weakly supervised learning approaches for remote sensing
- ✓ Created efficient deep learning models using reinforcement learning for adaptive computation
- ✓ Built ML models for sustainability applications including poverty mapping and farmland delineation

Computer Vision Engineer

Planet Labs

📅 June 2017 – July 2018
 📍 San Francisco, CA

- ✓ Built large-scale object detection dataset with **100K+** annotated satellite images
- ✓ Improved small object detection accuracy in low-resolution aerial imagery using convolutional detectors
- ✓ Conducted research to tackle unique challenges of satellite image object detection

Computer Vision Engineer

Autel Robotics

Manager: Youngwoo Seo, Ph.D.

📅 August 2016 – June 2017
 📍 San Ramon, CA

- ✓ Designed long-term target following system for next-generation drones
- ✓ Implemented online learning method for real-time single object tracking on embedded platforms
- ✓ Deployed and optimized tracking algorithms on low-end embedded systems

Computer Vision Algorithm Engineer Intern

Futurewei Technologies (Huawei R&D)

Manager: Dong-Qing Zhang, Ph.D.

📅 November 2015 – May 2016
 📍 Bridgewater, NJ

- ✓ Designed subspace learning method for stranger detection in family photo albums using deep CNNs
- ✓ Developed probabilistic graph-based approach for semantic role assignment in family photos

🎓 EDUCATION

Rochester Institute of Technology

Ph.D. in Imaging Science, Chester F. Carlson Center for Imaging Science

Thesis: Aerial visual vehicle detection and tracking using an adaptive, multi-modal sensor

Advisor: Matthew J. Hoffman, Ph.D. — Co-Advisors: Anthony Vodacek, Ph.D., John Kerekes, Ph.D.

📅 August 2011 – May 2016

University of Bridgeport

M.S. in Electrical Engineering

Thesis: Environmental non-speech sound classification with a new set of time-domain features

Advisor: Buket D. Barkana, Ph.D.

📅 August 2009 – May 2011

Eskisehir Osmangazi University

B.S. in Electrical and Electronics Engineering

Thesis: Autonomous parallel parking of non-holonomic vehicles

Advisor: Osman Parlaktuna, Ph.D.

📅 September 2004 – May 2009

RESEARCH EXPERIENCE

Graduate Research Assistant

Rochester Institute of Technology, Chester F. Carlson Center for Imaging Science
 Advisor: *Matthew J. Hoffman, Ph.D.*

 April 2012 – July 2016

 Rochester, NY

- ✓ Conducted research on aerial vehicle detection and tracking using adaptive, multi-modal sensors
- ✓ Developed computer vision and ML methods for vehicle detection, association, and tracking in aerial video
- ✓ Addressed challenges of medium-to-high altitude tracking through efficient use of hyperspectral data

Graduate Research Assistant

Rochester Institute of Technology
 Advisor: *Elizabeth Cherry, Ph.D.*

 May 2014 – June 2014

 Rochester, NY

- ✓ Performed research on 3-D MRI cardiac segmentation for understanding ventricular fibrillation mechanics

Graduate Research Assistant

University of Bridgeport, Electrical Engineering Department
 Advisor: *Buket D. Barkana, Ph.D.*

 August 2009 – May 2011

 Bridgeport, CT

- ✓ Conducted research on environmental sound classification for abnormal event detection
- ✓ Proposed novel pitch range-based features to improve classification accuracy

TEACHING EXPERIENCE

Teaching Assistant – Data for Development

Stanford University — Instructors: David Lobell, Marshall Burke, Stefano Ermon

Sept 2019 – Dec 2019

Graduate Teaching Assistant – Pattern Recognition

Rochester Institute of Technology — Instructor: John Kerekes, Ph.D.

Jan 2015 – May 2015

Graduate Teaching Assistant – Programming for Imaging Science

Rochester Institute of Technology — Instructor: Jeff Pelz, Ph.D.

Aug 2011 – May 2012

Graduate Teaching Assistant – Digital Image/Audio/Speech Signal Processing

University of Bridgeport — Instructor: Buket D. Barkana, Ph.D.

Aug 2010 – May 2011

Graduate Teaching Assistant – Non-linear Control Systems

Eskisehir Osmangazi University — Instructor: Osman Parlaktuna, Ph.D.

Jan 2009 – May 2009

TECHNICAL SKILLS

Machine Learning: PyTorch, TensorFlow, Caffe, Transformers, Multi-modal Learning, Computer Vision, LLMs

Programming: Python, C/C++, MATLAB, Shell Scripting, Git, HTML

Frameworks & Tools: OpenCV, Linux, Docker, AWS, IDL/ENVI, LaTeX

Research Areas: Video Understanding, Multi-modal ML, Self-Supervised Learning, Object Detection & Tracking

REFEREED JOURNAL PUBLICATIONS

- [1] E. Velterop, **B. UzKent**, J. Suckale, “Safe Shelter: A Case for Prioritizing Housing Quality in Climate Adaptation Policy by Remotely Sensing Roof Tarps in the San Francisco Bay Area”, *Earth’s Future* 10, no. 8 (2022) [\[PDF\]](#)
- [2] **B. UzKent**, A. Rangnekar, M.J. Hoffman, “Tracking in Aerial Hyperspectral Videos Using Deep Kernelized Correlation Filters”, *IEEE Transactions on Geoscience and Remote Sensing*, 57(1): 449-461, August 2018. [\[Code\]](#) [\[PDF\]](#)
- [3] **B. UzKent**, M.J. Hoffman, A. Vodacek, “Integrating Hyperspectral Likelihoods in a Multi-dimensional Assignment Algorithm for Aerial Vehicle Tracking”, *IEEE Journal of Selected Topics in Remote Sensing and Observation*, 9(9): 4325-4333, May 2016. [\[PDF\]](#)
- [4] **B. UzKent**, M.J. Hoffman, A. Vodacek, B. Chen, “Feature Matching with an Adaptive Optical Sensor in a Ground Target Tracking System”, *IEEE Sensors Journal*, 15(1): 510-519, January 2015. [\[PDF\]](#)
- [5] **B. UzKent**, B.D. Barkana, H. Cevikalp, “Non-speech environmental sound classification using SVMS with a new set of features”, *International Journal of Innovative Computing, Information and Control*, 8(5): 3511-3524, May 2012. [\[PDF\]](#)
- [6] B.D. Barkana, **B. UzKent**, I. Saricicek, “Normal and abnormal non-speech audio event detection using MFCC and PR-based feature sets”, *Advanced Materials Research*, Volume 601, pp: 200–208, December 2012.
- [7] B.D. Barkana, **B. UzKent**, I. Saricicek, “Environmental noise classifier using a new set of feature parameters based on pitch range”, *Applied Acoustics*, 72(11): 841–848, November 2011. [\[PDF\]](#)
- [8] **B. UzKent**, B.D. Barkana, J. Yang, “Automatic environmental noise source classification model using fuzzy logic”, *Expert Systems with Applications*, 38(7): 8751–8755, July 2011. [\[PDF\]](#)

REFEREED CONFERENCE PUBLICATIONS

- [1] T. Poppi, **B. UzKent**, A. Garg, L. Porto, G. Kessler, Y. Yang, M. Cornia, L. Baraldi, R. Cucchiara, F. Schiffrers, “CounterVid: Counterfactual Video Generation for Mitigating Action and Temporal Hallucinations in Video-Language Models”, *Conference on Empirical Methods in Natural Language Processing*, **EMNLP-26**, 2026. [\(Main Track\)](#) [\[PDF\]](#)
- [2] G. Sun, A. Singhal, **B. UzKent**, M. Shah, C. Chen, G. Kessler, “From Frames to Clips: Efficient Key Clip Selection for Long-Form Video Understanding”, *European Conference on Computer Vision Workshops*, **ECCVW-26**, 2026. [\[PDF\]](#)
- [3] R. Jain, K. Doshi, **B. UzKent**, G. Kessler, “Narrative Aligned Long Form Video Question Answering”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, **CVPRW-26**, 2026. (Best Paper Candidate) [\[PDF\]](#)
- [4] J. Yi, **B. UzKent**, O. Ignat, Z. Li, A. Garg, X. Yu, L. Liu, “Augment the Pairs: Semantics-Preserving Image-Caption Pair Augmentation for Grounding-Based Vision and Language Models”, *IEEE Winter Conference on Applications of Computer Vision*, **WACV-24**, January 2024. [\[PDF\]](#)
- [5] K. Doshi, A. Garg, **B. UzKent**, X. Wang, “A Multimodal Benchmark and Improved Architecture for Zero Shot Learning”, *IEEE Winter Conference on Applications of Computer Vision*, **WACV-24**, January 2024. [\[PDF\]](#)
- [6] **B. UzKent**, A. Garg, W. Zhu, K. Doshi, J. Yi, X. Wang, M. Omar, “Dynamic Inference with Grounding Based Vision and Language Models”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, **CVPR-23**, pp. 2624-2633, 2023. [\[PDF\]](#)
- [7] S. Gao, **B. UzKent**, Y. Shen, H. Huang, H. Jin, “Learning to Jointly Share and Prune Weights for Grounding Based Vision and Language Models”, *International Conference on Learning Representations*, **ICLR-23**, May 2023. [\[PDF\]](#)

- [8] M. Yin, **B. UzKent**, Y. Shen, H. Jin, “GOHSP: A Unified Framework of Graph and Optimization-based Heterogeneous Structured Pruning for Vision Transformer”, *AAAI Conference on Artificial Intelligence*, **AAAI-23**, February 2023. [\[PDF\]](#)
- [9] Q. Lu, Y.C. Shu, **B. UzKent**, T. Hua, Y. Shen, H. Jin, “Lite-MDETR: A Lightweight Multi-Modal Detector”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, **CVPR-22**, pp. 12206-12215, 2022. [\[PDF\]](#)
- [10] S. Chakraborty, **B. UzKent**, K. Ayush, K. Tanmay, E. Sheehan, S. Ermon, “Efficient Conditional Pre-training for Transfer Learning”, *CVPR Workshop*, **CVPRW-22**, pp. 4241-4250, 2022. [\[PDF\]](#)
- [11] **B. UzKent**, K. Ayush, C. Meng, M. Burke, D. Lobell, S. Ermon, “Geography-Aware Self-Supervised Learning”, *IEEE/CVF International Conference on Computer Vision*, **ICCV-21**, pp. 10181-10190, 2021. [\[Code\]](#) [\[PDF\]](#)
- [12] K. Ayush, A. Sinha, J. Song, **B. UzKent**, H. Jin, S. Ermon, “Negative Data Augmentation”, *International Conference on Learning Representations*, **ICLR-21**, May 2021. [\[Code\]](#) [\[PDF\]](#)
- [13] **B. UzKent**, K. Ayush, M. Burke, D. Lobell, S. Ermon, “Efficient Poverty Mapping from High Resolution Remote Sensing Images”, *AAAI Conference on Artificial Intelligence*, **AAAI-21**, February 2021. [\[PDF\]](#)
- [14] J. Lee, D. Grosz, **B. UzKent**, S. Zheng, M. Burke, D. Lobell, S. Ermon, “Predicting Livelihood Indicators from Community-Generated Street-Level Imagery”, *AAAI Conference on Artificial Intelligence*, **AAAI-21**, February 2021. [\[Code\]](#) [\[PDF\]](#)
- [15] **B. UzKent**, K. Ayush, M. Burke, D. Lobell, S. Ermon, “Generating Interpretable Poverty Maps Using Object Detectors in Satellite Images”, *International Joint Conference on Artificial Intelligence*, **IJCAI-20**, pp. 4410-4416, August 2020. [\[PDF\]](#)
- [16] **B. UzKent**, S. Ermon, “Learning Where and When to Zoom Using Deep Reinforcement Learning”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, **CVPR-20**, pp. 12345-12354, June 2020. [\(Oral Presentation\)](#) [\[Code\]](#) [\[PDF\]](#)
- [17] H.L. Aung, **B. UzKent**, M. Burke, D. Lobell, S. Ermon, “Farmland Parcel Delineation using Spatio-temporal Convolutional Networks”, *CVPR Workshop on Agriculture-Vision*, **CVPRW-20**, pp. 76-77, June 2020. [\[Code\]](#) [\[PDF\]](#)
- [18] **B. UzKent**, C. Yeh, S. Ermon, “Efficient Object Detection in Large Images Using Deep Reinforcement Learning”, *IEEE Winter Conference on Applications of Computer Vision*, **WACV-20**, pp. 1824-1833, March 2020. [\[Code\]](#) [\[PDF\]](#)
- [19] V. Sarukkai, A. Jain, **B. UzKent**, S. Ermon, “Cloud Removal from Satellite Images using Spatiotemporal Generative Networks”, *IEEE Winter Conference on Applications of Computer Vision*, **WACV-20**, pp. 1796-1805, March 2020. [\[Code\]](#) [\[PDF\]](#)
- [20] **B. UzKent**, E. Sheehan, C. Meng, D. Lobell, M. Burke, S. Ermon, “Learning to Interpret Satellite Images using Wikipedia”, *International Joint Conference on Artificial Intelligence*, **IJCAI-19**, pp. 3620-3626, July 2019. [\[Code\]](#) [\[PDF\]](#)
- [21] E. Sheehan, C. Meng, M. Tan, **B. UzKent**, N. Jean, D. Lobell, M. Burke, S. Ermon, “Predicting Economic Development using Geolocated Wikipedia Articles”, *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, **KDD-19**, pp. 2698-2706, July 2019. [\[Code\]](#) [\[PDF\]](#)
- [22] **B. UzKent**, Y. Seo, “EnKCF: Ensemble of Kernelized Correlation Filters for High-Speed Object Tracking”, *IEEE Winter Conference on Applications of Computer Vision*, **WACV-18**, pp. 1133-1141, March 2018. [\[Code\]](#) [\[PDF\]](#)
- [23] **B. UzKent**, A. Rangnekar, M.J. Hoffman, A. Vodacek, “Aerial Vehicle Tracking by Adaptive Fusion of Likelihood Maps”, *IEEE Workshop on Perception Beyond the Visible Spectrum, CVPR Workshop*, **CVPRW-17**, pp. 39-48, July 2017. [\[PDF\]](#)

- [24] **B. Uz Kent**, M.J. Hoffman, A. Vodacek, “Real time Target Detection and Tracking in Aerial Video using Hyperspectral Features”, *IEEE Workshop on Moving Cameras Meet Video Surveillance, CVPR Workshop, CVPRW-16*, pp. 36-44, June 2016. [\[PDF\]](#)
- [25] **B. Uz Kent**, M.J. Hoffman, A. Vodacek, “Spectral Validation of Measurements in a Vehicle Tracking DDDAS”, *International Conference on Computational Science*, Volume 51, pp. 2493-2502, June 2015. [\[PDF\]](#)
- [26] **B. Uz Kent**, M.J. Hoffman, A. Vodacek, “Background Image Understanding and Adaptive Imaging for Vehicle Tracking”, *SPIE Airborne Intelligence, Surveillance, Reconnaissance (ISR) Systems and Applications XII*, pp. 94600-94607, April 2015. [\[PDF\]](#)
- [27] **B. Uz Kent**, M.J. Hoffman, A. Vodacek, “Efficient Integration of Spectral Features for Vehicle Tracking utilizing an Adaptive Sensor”, *SPIE Video Surveillance and Transportation Imaging Applications*, pp. 940707-940717, February 2015. [\[PDF\]](#)
- [28] **B. Uz Kent**, M.J. Hoffman, E. Cherry, N. Cahill, “3-D MRI Cardiac Segmentation using Graph Cuts”, *IEEE Western NY Image Processing Workshop*, pp. 47-51, November 2014. [\[PDF\]](#)
- [29] **B. Uz Kent**, M.J. Hoffman, A. Vodacek, J.P. Kerekes, B. Chen, “Feature matching and adaptive prediction models in an object tracking DDDAS”, *Procedia Computer Science*, Volume 18, pp. 1939-1948, 2013. [\[PDF\]](#)
- [30] **B. Uz Kent**, B.D. Barkana, “Pitch range-based feature extraction for audio surveillance systems”, *IEEE International Conference on Information Technology: New Generations (ITNG)*, pp. 476-480, April 2011. [\[PDF\]](#)
- [31] B.D. Barkana, I. Saricicek, **B. Uz Kent**, “Performances of the ANN, SVM, K-means clustering methods recognizing different environmental sounds”, *24th European Conference on Operational Research*, Lisbon, Portugal, July 2010.
- [32] **B. Uz Kent**, O. Parlaktuna, “Autonomous parallel parking of non-holonomic vehicles”, *13th National Conference in Middle East Technical University*, Ankara, Turkey, 2009.

🕒 PAPERS UNDER REVIEW

- [1] A. Blume, **B. Uz Kent**, S. Chaudhuri, G. Kessler, “Learning to Rank Caption Chains for Video-Text Alignment”, *European Conference on Computer Vision, ECCV-26*. [\[PDF\]](#)

🔍 PROFESSIONAL SERVICE

Peer Reviewer: IEEE TGRS, IEEE TIFS, NeurIPS, WACV, ICCV, BMVC, IEEE TIP, ICML, ICLR, CVPR, IEEE Sensors Journal, Nature Machine Intelligence, IEEE Access

HONORS & AWARDS

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|--|-------------------------------|
| ★ Amazon Invention Award | June 2024 |
| ★ RIT Graduate Scholarship Award | September 2011 – May 2016 |
| ★ University of Bridgeport Dean’s Scholarship Award | August 2009 – May 2011 |
| ★ University of Bridgeport Outstanding Student Award | May 2011 |
| ★ Fulbright Opportunity Grant | August 2009 |
| ★ Erasmus Exchange Student | September 2007 – January 2008 |

LANGUAGES

- | | |
|--|---|
|  English – Advanced/Fluent |  Turkish – Native |
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